

Credit Risk Default Prediction

Executive Summary

Home Credit Default Risk Dataset · 307,511 applicants · 122 features

ANALYTICAL REPORT

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github.com/GaudiHan/Credit-Risk-Default-Prediction

0.753

ROC-AUC
XGBOOST MODEL

307K

LOAN APPLICATIONS
ANALYSED

8.1%

PORTFOLIO DEFAULT
RATE (BASELINE)

3.5×

RISK MULTIPLIER
LOW CREDIT SCORE

KEY FINDINGS

- External credit scores are the dominant predictor.** `EXT_SOURCE_2` is the #1 feature. Applicants scoring below `0.3` have a `15.88%` default rate — **3.5× higher** than those above `0.6` (`4.56%`).
- Debt burden shows non-linear risk.** The highest default rate (`8.81%`) occurs in the `2x–3.5x` credit-to-income band — not at the highest debt levels — suggesting compensating factors in very high DTI applicants.
- Employment tenure is a material risk factor.** Borrowers employed less than one year default at `10.98%` vs. `6.41%` for those employed over 5 years — a **1.7× risk multiplier**.
- XGBoost achieves 0.753 ROC-AUC.** At 80% recall threshold, precision reaches 13.6% — nearly double the base default rate, making credit review 2× more efficient.

RISK SEGMENTATION

SEGMENT	DEFAULT RATE	RISK LEVEL
<code>EXT_SOURCE_2 < 0.3</code> (Low Credit Score)	15.88%	High
<code>Employed < 1 year</code>	10.98%	High
<code>Credit/Income 2x–3.5x</code> (Medium DTI)	8.81%	Medium
<code>Employed > 5 years</code>	6.41%	Low
<code>EXT_SOURCE_2 > 0.6</code> (High Credit Score)	4.56%	Low

METHODOLOGY

DATA	307K applications + bureau history. Missing values imputed; class imbalance handled via <code>class_weight='balanced'</code> .
FEATURES	4 engineered features (debt burden, repayment affordability, employment stability, bureau exposure) added to 122 base features.
MODEL	<code>XGBoost</code> with hyperparameter tuning (RandomizedSearchCV). Achieved <code>ROC-AUC 0.753</code> — captures 80% of defaults at 13.6% precision.
EXPLAIN	<code>SHAP</code> TreeExplainer identifies <code>EXT_SOURCE_2</code> as the dominant predictor. <code>Feature importance</code> available for stakeholders.

RECOMMENDATIONS

IMMEDIATE

Prioritise Low Credit Score Flagging

Applicants with `EXT_SOURCE_2` below `0.3` show `15.88%` default rates. Implement enhanced review or additional verification for this segment.

MEDIUM-TERM

Deploy Model Score as Decision Supplement

Integrate Random Forest default probability (ROC-AUC `0.737`) alongside bureau checks. Target high-recall threshold to capture 80% of defaults.

STRATEGIC

Investigate Medium DTI Segment

The `2x–3.5x` credit-to-income band shows peak default risk (`8.81%`). Further segmentation analysis could uncover specific risk drivers within this group.